

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Library Management System

1. Introduction

1.1 Purpose

The purpose of the Library Management System is to automate the activities of a library such as adding books, registering members, issuing books, returning books, searching books, and maintaining library records.

1.2 Scope

The Library Management System provides an easy and efficient way to manage library operations. It reduces manual work and helps the librarian maintain accurate records of books and members.

1.3 Objectives

- To maintain details of books and members.
- To provide easy book searching.
- To manage book issue and return operations.
- To calculate fines for late returns.
- To maintain accurate library records.
- To reduce paperwork and manual errors.
- To save time for librarians and students.

2. Overall Description

2.1 Product Perspective

The Library Management System is a computerized system that replaces the traditional manual library management process. The system stores library information in a database and allows authorized users to access and update it.

2.2 Product Functions

The system shall provide the following functions:

- User registration and login.
- Add, update and delete book details.
- Search and view books.
- Register and manage members.
- Issue books to members.
- Return books.
- Calculate overdue fines.
- Maintain transaction history.
- Generate reports.

2.3 User Classes

Librarian

- Login to the system.
- Manage books.
- Manage members.
- Issue and return books.
- Check fines.
- Generate reports.

Student/Member

- Login or access the library system.
- Search for books.
- View book availability.
- Check issued books and due dates.

Administrator

- Manage users and system settings.
- Maintain overall system information.

3. Functional Requirements

FR1 – UserLogin

The system shall allow authorized users to log in using a username and password.

FR2 – BookManagement

The librarian shall be able to:

- Addnewbooks.
- Updatebookdetails.
- Deletebooks.
- Viewavailablebooks.

FR3 – BookSearch

The system shall allow users to search for books using:

- Booktitle
- Authorname
- BookID
- Category

FR4 – MemberManagement

The librarian shall be able to add, update, view and remove member details.

FR5 – BookIssue

The system shall allow the librarian to issue an available book to a registered member and record the issue date and due date.

FR6 – BookReturn

The system shall record the return of books and update the availability status.

FR7 – Fine Calculation

The system shall calculate the fine when a book is returned after its due date.

FR8 – Reports

The system shall generate reports such as:

- Available books
- Issued books
- Returned books
- Overdue books
- Member details
- Fine details

4. Non-Functional Requirements

4.1 Performance

- The system should respond quickly to user requests.
- Searching and retrieving book information should be fast.
- The system should support multiple users when required.

4.2 Security

- Only authorized users should access the system.
- Passwords should be protected.
- Users should have access only to the functions permitted for their role.

4.3 Reliability

- The system should maintain accurate records.
- Data should not be lost during normal operation.
- Regular database backup should be supported.

4.4 Usability

- The user interface should be simple and easy to understand.
- The system should provide clear error messages.

- Users should be able to perform common operations easily.

4.5 Maintainability

- The software should be easy to maintain and update.
- Changes to one module should not affect other modules unnecessarily.

4.6 Availability

- The system should be available during library working hours.
- The database should be accessible whenever the system is running.

5. External Interface Requirements

5.1 User Interface

The system should provide:

- Login page
- Dashboard
- Book management page
- Member management page
- Issue/return page
- Search page
- Reports page

5.2 Hardware Interface

The system may require:

- Computer or laptop
- Keyboard and mouse
- Monitor
- Printer, if reports are printed
- Server/database system, if deployed over a network

5.3 Software Interface

The system may use:

- Operating System: Windows/Linux
- Database: MySQL
- Front End: HTML, CSS, JavaScript
- Back End: Java/Python/PHP or another suitable programming language

6. System Constraints

- Only authorized users can access administrative functions.
- A book can be issued only if it is available.
- A member must be registered before issuing a book.
- The system depends on the availability of the database.
- Fine rules must follow the library's policies.

7. Database Requirements

The system may maintain the following tables:

Book

- Book_ID
- Book_Name
- Author
- Publisher
- Category
- Quantity
- Availability_Status

Member

- Member_ID
- Member_Name
- Department
- Phone_Number
- Email

Issue/Return

- Transaction_ID

- Book_ID
- Member_ID
- Issue_Date
- Due_Date
- Return_Date
- Fine

User

- User_ID
- Username
- Password
- Role

8. Use Case Requirements

The major use cases are:

1. Login
2. Manage Books
3. Search Books
4. Manage Members
5. Issue Book
6. Return Book
7. Calculate Fine
8. Generate Reports
9. Logout

Main Actors

- Librarian
- Student/Member
- Administrator

9. Assumptions and Dependencies

Assumptions

- Users have basic knowledge of computers.

- The librarian maintains correct book and member information.
- Each book and member has a unique ID.

Dependencies

- The system depends on the database for storing information.
- The system requires proper hardware and software configuration.
- Network connectivity may be required for a multi-user system.

10. Conclusion

The Library Management System provides an efficient and organized method for managing library activities. It automates book management, member management, issue and return operations, fine calculation and report generation. The system reduces manual effort, improves accuracy and makes library operations faster and easier.